UNIT 6: APPLICATIONS OF INTEGRATION

**Include: accumulation, linear motion, area between curves, volume, cross section, arc length**

**<u>Accumulation</u>**

Integration can be used to find how much something has changed during a specific interval, but without additional information, you wouldn't be able to find where that thing ends up.
For example, let's say you have the velocity function for a particle (which is the derivative of the position function), and you want to find out how much the particle traveled between times t =1 and t = 4. With a simple integration from 1 to 4, you can figure that out.

But what if you want to know **exactly what position the particle is in at t=4**? You'd have to know the **starting position** of the particle and **add the distance traveled** to find where it ended up. That's what accumulation is.

$$f(b) = f(a) + \int_a^b f'(x)\,dx$$

Essentially, this means that:

**What I have now = Starting value + accumulation since starting value**

Here's a quick example:

*A particle moves along a line so that its velocity for t >= 0 is given by $v(t) = 6x^2$. If the particle's position at time t=2 is 3, what is the position of the particle at t=7?*

$x(7) = x(2) + \int_2^7 v(t)\,dt$

**<u>Linear Motion</u>**

$$\text{Displacement} = \int_a^b V(t)\,dt$$

**Displacement** - position shift

Displacement can be found by simply integrating the velocity. The negative areas below the x-axis subtract from total displacement.

**Distance traveled** - total distance traveled

To find the total distance traveled, find where the particle changes direction (velocity function changes sign) and integrate in pieces. This is the same as integrating the absolute value of the velocity function. This is so the negative movement isn't subtracted from the total distance.

$$\text{Distance Traveled} = \int_a^b |V(t)|\,dt$$

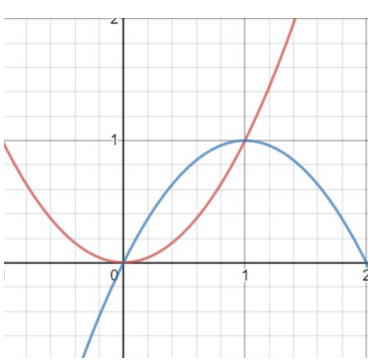
**<u>Area Between Curves</u>**

When you're finding the area between two curves, you are either solving in terms of x (dx) or in terms of y (dy). To determine which way to solve a problem, you consider factors like how the equations are presented and how they "stack" (more on that later). Let's start with dx.

Area $= \int_a^b$ *(upper curve - bottom curve) dx*

Here are the steps to solve for an area between two curves:

1. Sketch the graphs
2. Identify the closed region (Pay close attention! Sometimes there are multiple areas.)
3. Decide how to slice the region, vertically or horizontally
    a. Vertically → dx
    b. Horizontally → dy
4. Find intervals for integration
5. Set up the equation for the height of the area and integrate




Here's an example:

*Solve for the area between $y = x^2$ and $y = 2x - x^2$.*

Sketch the graphs and see where the area is bound. Note that the graphs intersect at x = 0 and x = 1. This is important because it gives us the limits for our integration.
Now to set up our integration, we need to identify which curve is the top curve and which is the bottom curve. Visually, you can see the blue curve ($2x - x^2$) is on top of the red curve ($x^2$).

$\int_0^1 [ (2x - x^2) - (x^2) ] \, dx$ is the correct integral for this problem.

***Sidenote: AREA WILL ALWAYS BE POSITIVE. If your final answer isn't positive, then it's not correct.***

Okay, dy problems aren't that different.

Area $= \int_c^d$ *(right curve - left curve) dy*

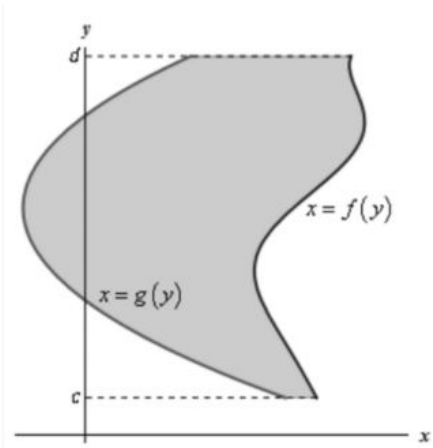
Now, instead of top minus bottom (bigger y value minus smaller y value), you're doing right minus left (bigger x value minus smaller x value). Your limits are now in terms of y instead of x. Makes sense, right? From there, the process is basically the same.

You might be saying: Woah, wait a minute. Why would we choose to solve in terms of y?

There's a few different reasons.

Firstly, if the equations $x = y^2 - 3y$ and $x = 3 - y^2$ are given to you, it may be in your best interests to keep those equations that way instead of setting them equal to y. Some equations may even be (basically) impossible to set equal to y.

Secondly, look at these two curves. If we tried to slice it vertically, we'd run into all kinds of troubles. On the left side, we'd be subtracting the same equation from itself. On the right side, it would split into two different "top minus bottom" situations, one of which is also an equation subtracting itself.

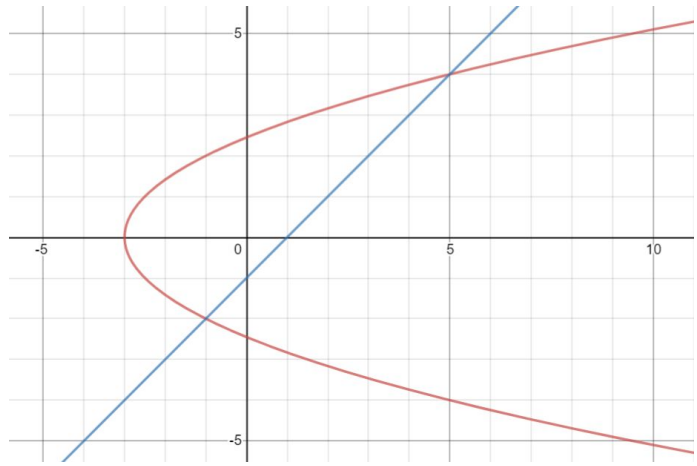


In this case, we *must* slice it horizontally and solve it in terms of y. Trust me though-- it isn't that scary!

Here's an example:

*Solve for the area between* $x = \frac{1}{2} y^2 - 3$ *and* $y = x - 1$.

Sketch the graphs and see where the area is bound.




Now when you solve for intercepts to find your limit bounds, remember that since you're solving in terms of y, you're looking for the y values where they intersect. For this example, they're y = -2 and y = 4.

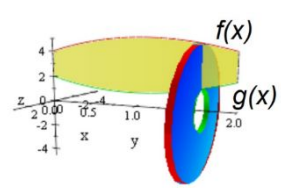
Then to set up the integration, identify the left and right curves and you're all set!

$\int_{-2}^{4} [ ( y + 1 ) - ( \frac{1}{2} y^2 - 3 ) ] \, dy$ is the correct integral for this problem. Remember that it is **right minus left**.

**<u>Volumes of Revolutions</u>**

Imagine that you take an area bound between two curves and rotate it around an axis. How would you find the volume of that now 3-dimensional shape? There are two methods we use to solve for the volume: the **Washer Method** and the **Shell Method**.

Let's start with the Washer method.




Conceptually, the washer method looks at a slice of the rotated object, shown in blue on the right. It finds the area of the big circle (created by the top function, *f(x)*) and subtracts the area of the small circle (created by the bottom function, *g(x)*) in order to find the blue area. It repeats this process from one limit to the other,

$$V = \pi \int_a^b [(f(x))^2 - (g(x))^2] \, dx$$

$$V = \pi \int_a^b (R^2 - r^2) dx$$

$$V = \pi \int_c^d (R^2 - r^2) dy$$

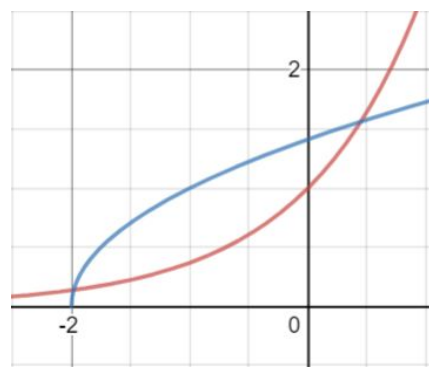
thus finding the volume of the entire rotated object. That's why the equation is basically finding the area of a circle: $\pi r^2$.

Now let's get into the nitty gritty details. Here are the steps to solving for the volume:

1. Draw a picture and identify the bounded region
2. The cross-sections must be <u>perpendicular</u> to the axis of rotation (which will be stated in the problem)
    a. This means that if the object is being rotated around the x axis, it must be sliced vertically, and if the object is being rotated around the y axis, it must be sliced horizontally.
3. If the cross-sections are vertical: Top - bottom, x-limits, dx
4. If the cross-sections are horizontal: Right - left, y-limits, dy

Here are some examples:

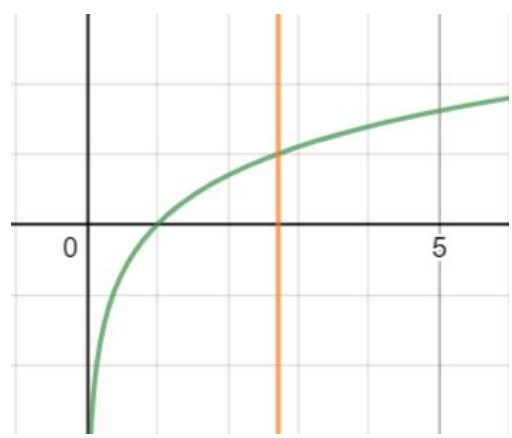
*Example 1: Find the volume when the region bounded by* $y = e^x$ *and* $y = \sqrt{x+2}$ *is rotated about the x-axis. (Calculator active)*




Because the graph is being rotated about the x-axis, the area is being sliced vertically. Using a calculator, find the points where the graphs intersect. Then identify the top curve (big R) and bottom curve (little r) and set up your integral:

$$\pi \int_{-1.981}^{0.448} (\sqrt{x+2})^2 - (e^x)^2 \, dx$$

*Example 2: Find the volume generated by revolving, about the y-axis, the region in the first quadrant bounded by the graph of y = ln{x} and the line x = e.*




Because the graph is being rotated about the y-axis, the area is being sliced horizontally. We know that the lower y-bound will be y=0 because the region is in the first quadrant. We know the ln x and x = e intersect at (e, 1) (or we can do some quick math), and that is the upper bound. Then identify the right curve (big R) and left curve (little r) and set up your integral:

$$\pi \int_0^1 (e)^2 - (e^y)^2 \, dy$$

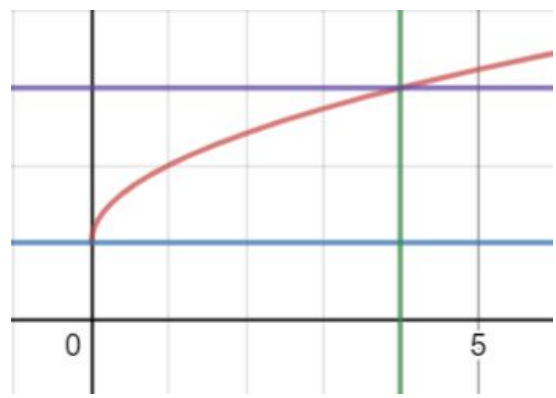
Okay-- but what if we want to rotate the region around a line that *isn't* an axis? That becomes a little trickier.

The main thing that changes is how you represent big R and little r. When we were revolving around an axis, R and r were just the equations. Now, with a different axis of rotation in the picture, they change. R becomes the difference between the axis of rotation and the curve furthest away from it, and r becomes the difference between the axis of rotation and the curve closer to it. Let's take a look at this example:

$$V = \pi \int_a^b (R^2 - r^2) dx$$

$$V = \pi \int_c^d (R^2 - r^2) dy$$

*Find the volume of the region enclosed by* $y = 1 + \sqrt{x}$ *and y = 1 and x = 4 that is rotated around the line y = 3.*




The purple line, y = 3, represents the axis of rotation in this picture. Since the axis of rotation is horizontal, that means the cross-sections will be vertical, and thus the problem will be dx.

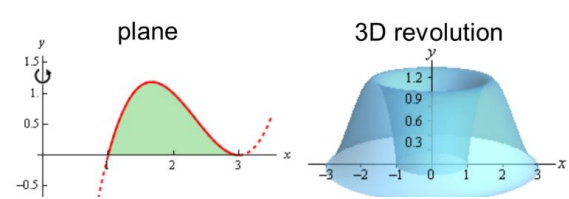
So what is the bigger radius R? You find the curve that's the furthest from the purple line, which is y = 1. In order to figure out what is subtracting what, we still follow the **top minus bottom** rule. Since y = 3 is the top and y = 1 is the bottom, big R is (3 - 1).

Finding smaller radius r is the same process. You find the curve that's closer to the axis of rotation, which is $y = 1 + \sqrt{x}$ . Following the top minus bottom rule, little r is (3 - ( 1 + $\sqrt{x}$ )). Make sense?

Go through the process of finding the limits. This should be your final integral:

$$\pi \int_0^4 (3-1)^2 - (3 - (1 + \sqrt{x}))^2 \, dy$$

If the axis of rotation had been underneath the bounded area, let's say y = -1, then big R would have been the red curve subtracting -1 and little r would have been the blue curve subtracting -1.

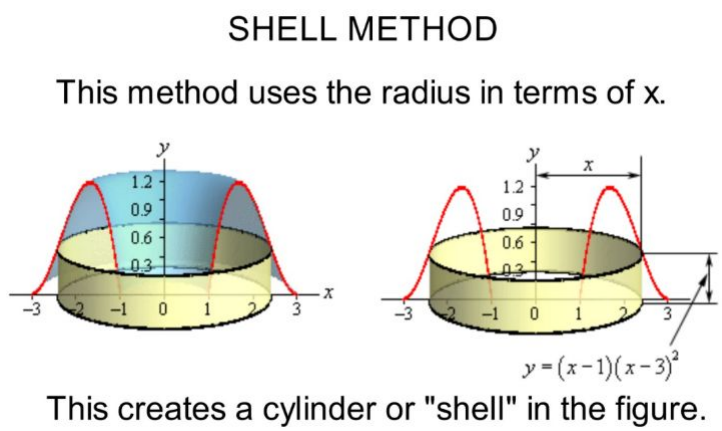


Okay, now let's talk about the Shell Method.

Contrary to the Washer Method that's chopping the object into thin circle rings, the Shell Method is like splitting the object into thin hollow cylinders. Imagine the rotated 3-dimensional object is an onion and you're shaving the outside layers off one at a time.

Look at the picture on the right. With the Washer method, since it's being rotated around the y-axis, the cross-sections would be horizontal. See why you can't do that? You'd end up subtracting the equation from itself. Your cross sections *must* be vertical in this scenario. That's where the Shell Method comes in-- <u>when the cross sections are **parallel** to the axis of rotation.</u>

Just like how the Washer Method was modeled after the circle, the Shell method is modeled after the cylinder. Since they are hollow cylinders, we're actually looking at the surface area. Imagine a rolled up sheet of paper. The circumference of the circle is $2\pi r$. When you unravel the sheet of paper, the height is the height of the cylinder, and the length is $2\pi r$. So, the surface area equation is $2\pi rh$. The Shell Method just uses r(x) and h(x) equations to represent the radius and height.

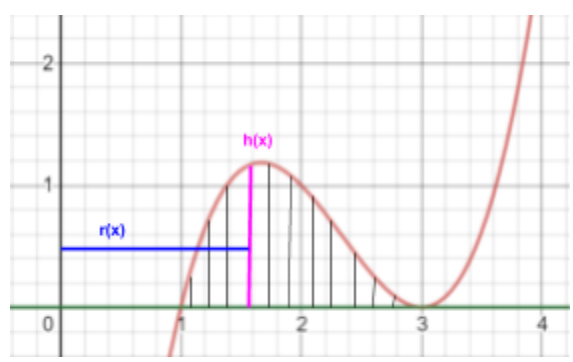



So, the equation for the Shell method is $2\pi\int_a^b r(x)\,h(x)\,dx$.

Here's our first example:

*Find the volume of the object when the area bounded by $y = (x - 1)(x - 3)^2$ and $y = 0$ is rotated about the y-axis.*

Since this area cannot be sliced horizontally, it must be sliced vertically and solved for in terms of x. The intersections are at x = 1 and x = 3.



The radius equation will be how far each cross section is from the axis of rotation. Since the axis of rotation is just the y-axis, the radius here is simply x.

The height equation will be the height of each cross-section, which is just the curve.

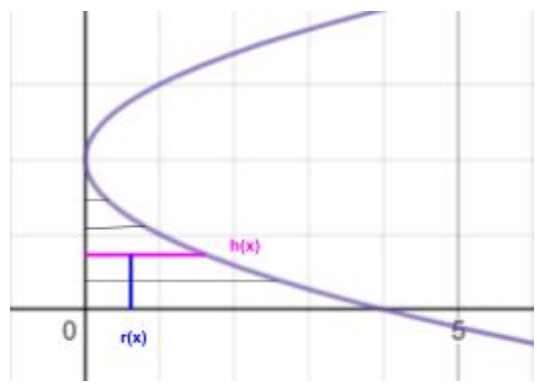
The final integral is $2\pi\int_1^3 x\,(x-1)(x-3)^2\,dx$. Not bad, huh?

Here's the next one:

*Find the volume when the area bounded by $x = (y - 2)^2$ in the first quadrant is rotated about the x-axis.*

It's worth addressing the fact that you could solve this with the Washer method, but since it's set up for you in terms of y, it's more convenient to stick with that.

Here, the r(x) is simply y and h(x) is the curve. Find what the y-limits are.




The final integral should be:

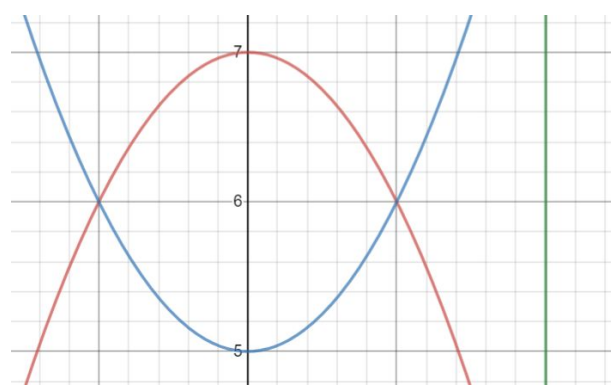
$$2\pi\int_0^2 y\,(y-2)^2\,dy$$

So what if the area isn't rotated around an axis? It isn't that big of a leap, don't worry. The r(x) equation just adjusts a little bit, either becoming (axis of rotation - x) or (x - axis of rotation), depending on which is top and which is bottom. Alternatively, it's either (axis of rotation - y) or (y - axis or rotation), depending on which is left and which is right. I'm telling you-- top minus bottom and right minus left are the golden rules for this concept.

Here's your example:

*Find the volume enclosed by the graphs $y = -x^2 + 7$ and $y = x^2 + 5$ rotated around the line $x = 2$.*

Sketch the two quadratics and draw the axis of rotation (the green line). Again, it's necessary to draw the cross-sections vertically in this case, otherwise you'd subtract the same equation from itself.



The height equation for the height of the cross sections will be the red curve minus the blue curve, as you can see. The radius equation will be (2 - x) since x = 2 is on the right. *Right minus left is key!*

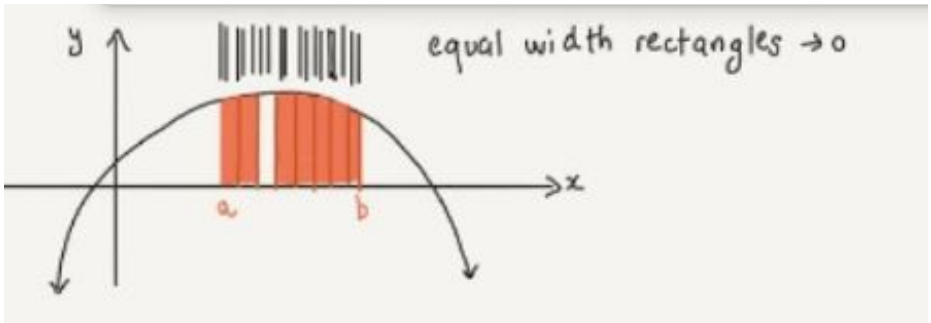
There will be more practice with these kinds of problems at the bottom of the page.

**<u>Cross Sections (Connor)</u>**

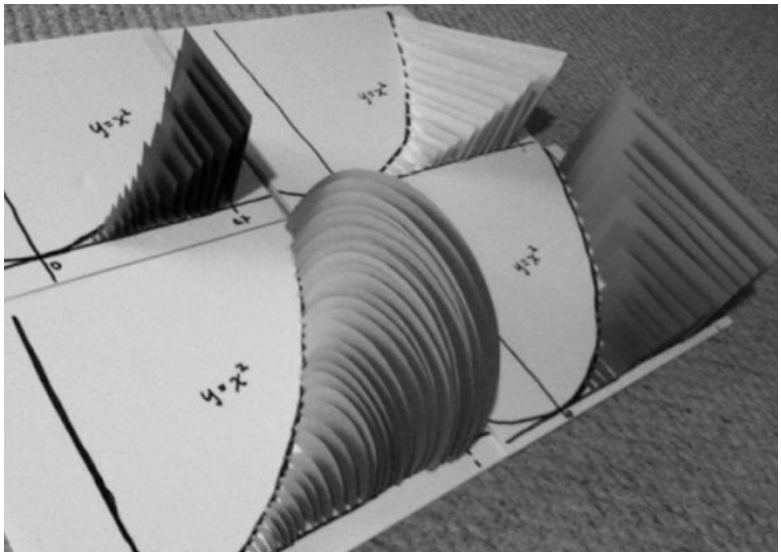
Let's get into cross sections!

**The Basics and Review**

We can use cross sections as another method to find the volume of a solid. Instead of rotating shapes around an axis, we can instead stack many 3D shapes against each other to find the volume of a shape. If we have a certain volume that we can't compute, we instead add up many shapes that we do know how to compute, much like normal accumulation. Compare the following:




*Accumulation for **area***



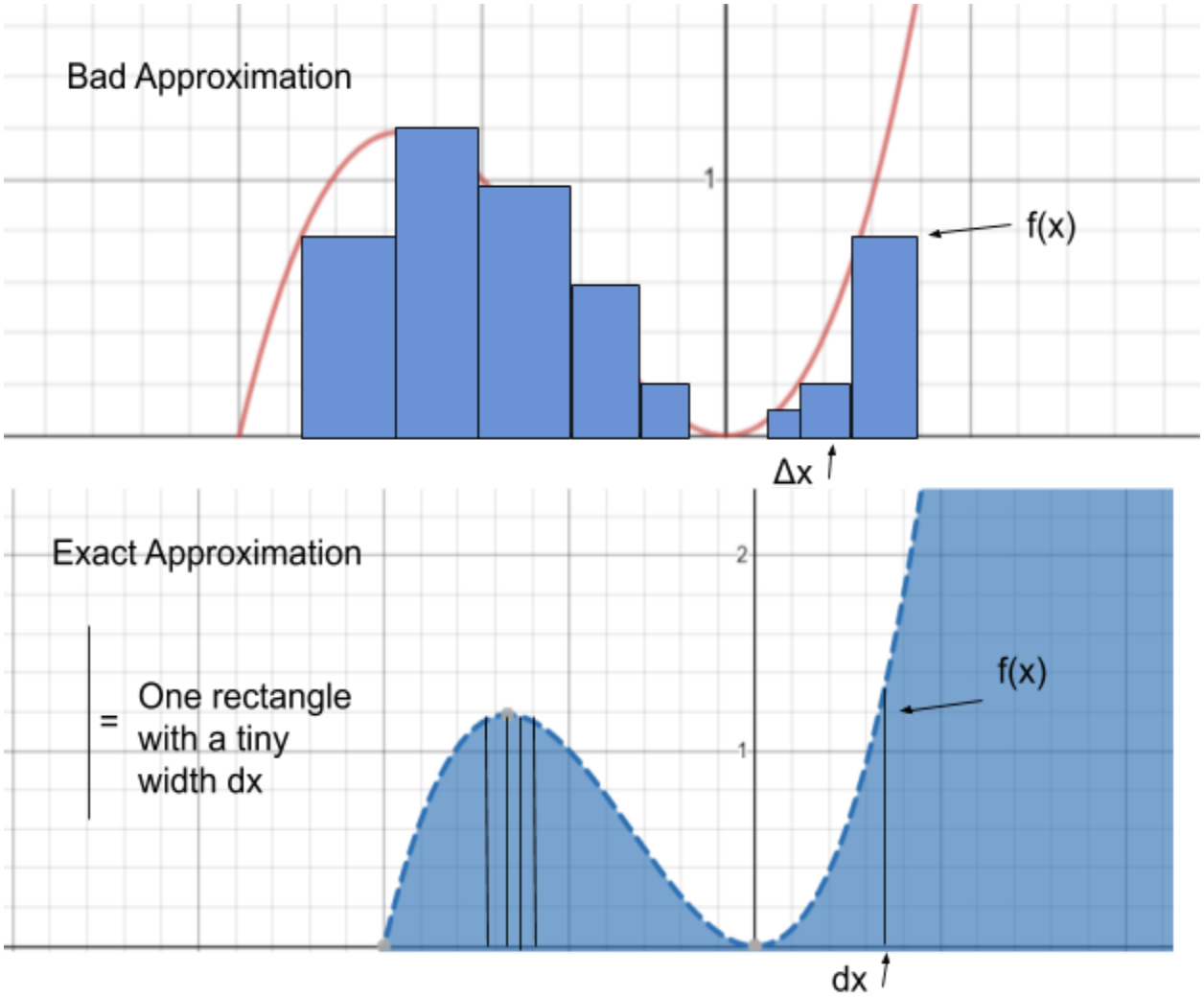

*Using accumulation of cross sections for* ***volume***

In the first picture, the area is found by adding together many 2D shapes.
In the second picture, the volume is found by adding together many 3D shapes(their width is very tiny).

Stop and think: how will we create these shapes? What will, for example, represent the width? How is this related to integration?

I know those are some tough questions. Let's break it down part by part. If you recall back to normal accumulation integration, the area under a curve is found by creating many rectangles and adding them

together. The more rectangles you include, the more accurate the area gets. See below:



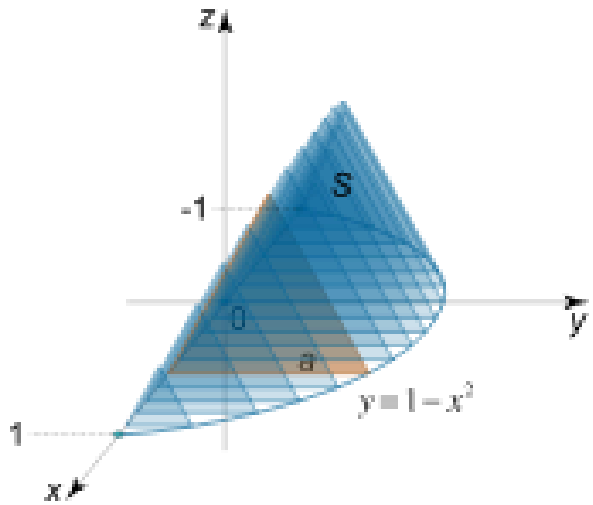

We start off with a finite number of rectangles each with a finite width Δx. Their length is the function value f(x). We then add them together to get a rough approximation (these are known as Riemann sums). This is easy to compute, because the area of a rectangle is simply length times width.

But we want something more accurate. So, through the limit process, we let Δx approach 0. When this happens, this sum turns into an integral. With an infinitesimal change in x, dx, the sum exactly approximates the area. I know what you are saying- I already know this!

Well, I felt it beneficial to review before we get into the next part.

**The Setup**

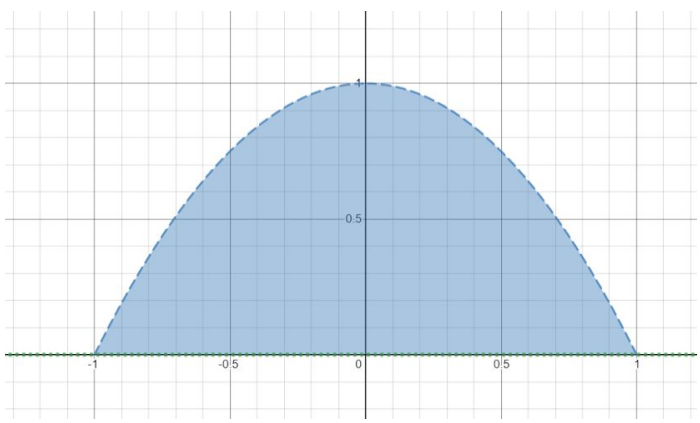
Imagine this 3D object.




*The base of a solid is bounded by the parabola y=1−x^2 and the x-axis. Find the volume of the solid if the cross sections are equilateral triangles perpendicular to the x−axis.*

How in the world are we going to find the area of this thing?

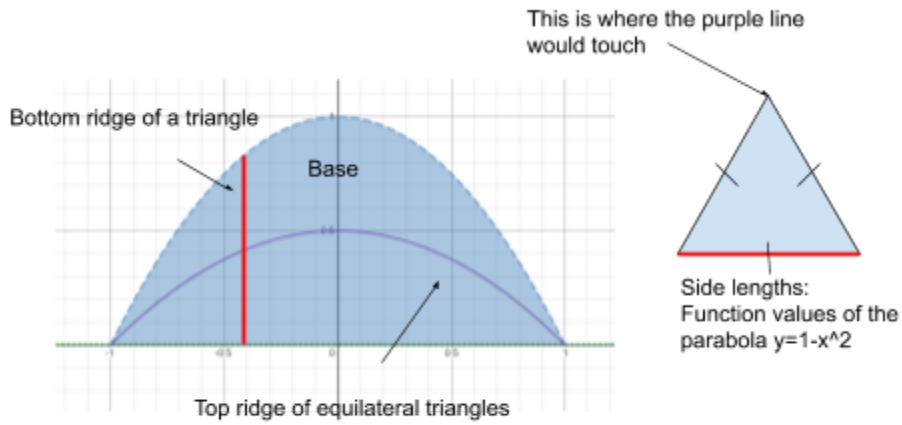
Let's begin by understanding conceptually how to use cross sections.Firstly, a problem will give you several bounds and functions to create the "base" of a 3D figure. In the case above, the base of the figure is bounded by y=1-x^2 and the x-axis. That looks like this:




Then, perpendicular to an axis given in the problem, we stack cross sections together to approximate the volume. This will determine whether to use dx or dy:

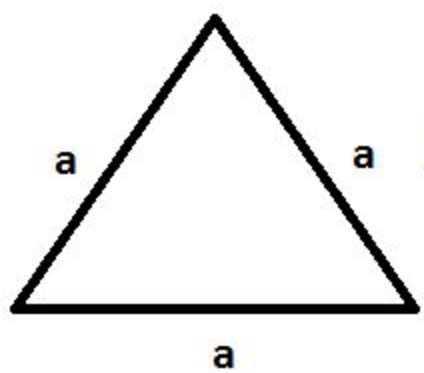
- use dx if we are perpendicular to the x-axis
- use dy if we are perpendicular to the y-axis

Now, let's get on to the problem. It calls for equilateral triangles perpendicular to the x-axis. To visualize this either refer to the image at the beginning, or look at my artistic interpretation below:



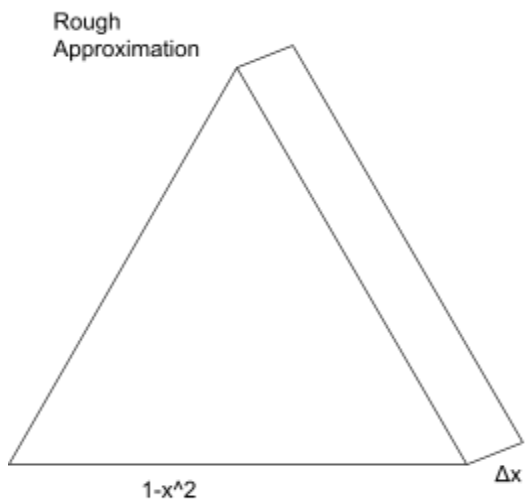

Imagine that these triangles are jumping out of the screen toward you, and the purple line outlines where their top points would be located.

To utilize these triangles, let's recall the equilateral triangle area formula.




$$A = \frac{\sqrt{3}}{4} a^2$$

As seen in the problem, the side *a* is represented by 1-x^2. Just like shown in 2D accumulation, I'll start off with a loose approximation that uses a few cross sections.

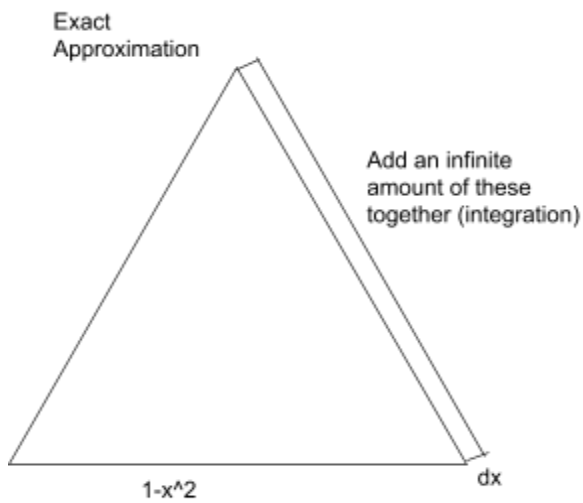



This 3D figure can be stacked against many others to formulate the volume. If I used 4 triangles, the math could look like this:

| | | | | | |
|---|---|---|---|---|---|
| x | -1 | -.5 | 0 | .5 | 1 |
| f(x) | 0 | .75 | 1 | .75 | 0 |

Volume of one triangular prism =((sqrt3)/4)a^2(Δx)
If we add together 4 and compute, we get:
Volume ≈ .5*((sqrt3)/4)(.75^2+1^2+.75^2+0^2) = 0.4600

Just like accumulation, if we, through the limit process, let Δx become dx, then we can evaluate the integral and find an even better volume!




$$V = \int_{-1}^{1} A(x)\,dx = \int_{-1}^{1} \frac{\sqrt{3}}{4}\left(1-x^2\right)^2 dx = \frac{\sqrt{3}}{4}\int_{-1}^{1}\left(1-2x^2+x^4\right)dx$$
$$= \frac{\sqrt{3}}{4}\left[x - \frac{2x^3}{3} + \frac{x^5}{5}\right]\Bigg|_{-1}^{1} = \frac{\sqrt{3}}{4}\left[\left(1-\frac{2}{3}+\frac{1}{5}\right) - \left(-1+\frac{2}{3}-\frac{1}{5}\right)\right]$$
$$= \frac{\sqrt{3}}{2}\left(1-\frac{2}{3}+\frac{1}{5}\right) = \frac{4\sqrt{3}}{15}.$$

(4sqrt3)/15=0.4618

Actually pretty close to our rough approximation!

This method could be used for any type of 2D cross section:

- Semi-circles
- Quarter-circles
- Squares

*With circles, make sure you use the correct value for the radius! If the circle's diameter is the entire length of the function's y values (or x values if you are integrating with dy), then the radius is one half, so you have to multiply the integral by ½. Also, for semicircles and quarter-circles, remember to multiply by ½ and ¼ respectively.

With that in mind, here are the general formulas:

Semicircles:

$$\frac{\pi}{8}\int_a^b [f(x)]^2 dx$$

(the radius is $\frac{f(x)}{2}$, so squaring that, we get $\frac{1}{4}$ which we pull out front and multiply by $\frac{\pi}{2}$ to get $\frac{\pi}{8}$)

Quarter-circles:

$$\frac{\pi}{4}\int_a^b [f(x)]^2 dx$$

(the radius is $f(x)$, so we don't have to pull anything out in front. Then we have $\frac{\pi}{4}$ because it's a quarter-circle).

**Resources:**

If you want more explanation, see Khan Academy:
https://www.khanacademy.org/math/ap-calculus-ab/ab-applications-of-integration-new/ab-8-8/v/volume-triangular-cross-section

If you want more practice, see here:
https://www.math24.net/volume-solid-with-known-cross-section/

**<u>Arc Length</u>**

Arc length is just the length of a curve. Here's the lovely equations for it.

**Definition of Arc Length**

Let the function $y = f(x)$ represent a smooth curve on the interval $[a, b]$. The **arc length** of $f$ between $a$ and $b$ is

$$s = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx.$$

Similarly, for a smooth curve $x = g(y)$, the **arc length** of $g$ between $c$ and $d$ is

$$s = \int_c^d \sqrt{1 + [g'(y)]^2}\,dy.$$

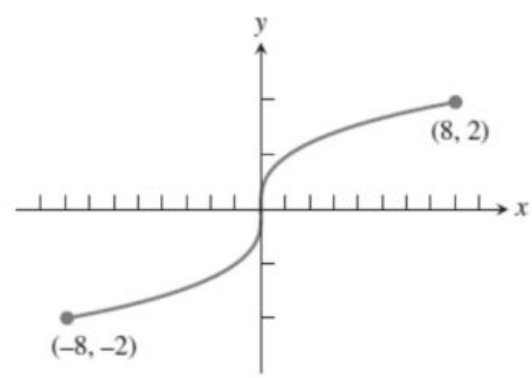
Pretty simple, yeah? Here are your examples:

Find the exact length of the following curve:

$y = \frac{4\sqrt{2}}{3}x^{3/2} - 1$ for $0 \le x \le 1$

$s = \int_0^1 \sqrt{1 + (2\sqrt{2}x^{1/2})^2}\,dx$ is the equation. You should be able to solve this by hand!

Find the length of the curve $y = x^{1/3}$ between (-8, -2) and (8, 2).

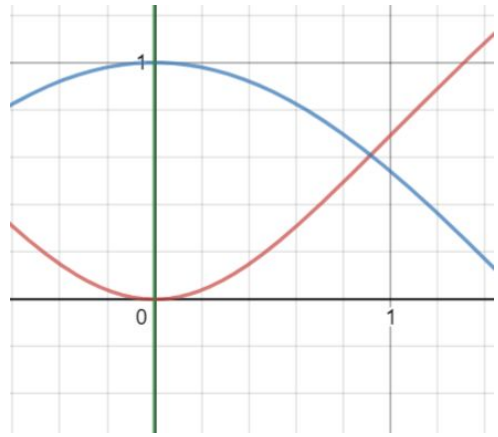



There's an issue with trying to solve this in terms of x. At x = 0, there is a vertical tangent line, so the derivative does not exist there. Since the arc length formula uses the derivative, this will cause issues for us.

Instead, you would solve this in terms of y by simply changing the equation to $x = y^3$ and solving in terms of y.

Lastly, let R be the region enclosed in the first quadrant by the graphs of $y = \ln(x^2 + 1)$, $y = \cos x$, and the y-axis. Write an expression involving one or more integrals that gives the length of the boundary of R.

This problem seems relatively easy. You're just finding the length of the sides of the enclosed region here in the first quadrant.




Using a calculator, find that intersection between the blue and red curves to get your limit. Then, draw up two separate integrals for them.

The biggest thing to emphasize here is *don't forget the vertical tidbit.* The green line is part of the region's boundary but it may be easy to forget it in the excitement of writing arc length equations.

Whew, okay! You've made it to the end. Here are some examples to help check your understanding:

First up, a calculator-active FRQ.

1. A particle moves along the y-axis with velocity given by $v(t) = t\,sin(t^2)$ for $t \geq 0$.
    a. In which direction (up or down) is the particle moving at time t = 1.5? Why?
    b. Find the acceleration of the particle at time t = 1.5. Is the velocity of the particle increasing at t = 1.5? Why or why not?
    c. Given that y(t) is the position of the particle at time *t* and that y(0) = 3, find y(2).
    d. Find the total distance traveled by the particle from t = 0 to t = 2.
2. Find the area of the region bounded by the graphs x = y(2 - y) and x = -y. (No calculator)
3. Region *R* is bounded by $y = x^2 - 3$, x = 6, and the x-axis. Find the volume when *R* is rotated around the line:
    a. y = 0
    b. y = -2
    c. x = 6
    d. x = 1
4. Let *R* be the region in the first quadrant bounded by the graph of $y = 3x - x^2$ and the x-axis.
    a. Find the area of R.
    b. A solid is generated when R is revolved around the line y = 6. Find its volume.
    c. A solid is generated when R is revolved around the line x = -2. Find its volume.

5. Find the volume of the solid whose base is the region bounded by the curves $y = x^2$ and $y = 8 - x^2$, and whose cross sections are perpendicular to the x-axis are
    a. Squares
    b. Semicircles
6. Let *R* be the region in the first quadrant bounded by the y-axis and the graphs $y = 4x - x^3 + 1$ and $y = \frac{3}{4}x$. Find the perimeter of the region *R*.